Calibration results
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Normalized Residuals

Reprojection error (cam0): mean 0.1700993760450453, median 0.12631178660313505, std: 0.1588506636665267
Reprojection error (cam1): mean 0.16657500500086836, median 0.12496487519285435, std: 0.15793375682231509
Gyroscope error (imu0): mean 2.504216230636108, median 2.019924041963815, std: 1.8557866101626517
Accelerometer error (imu0): mean 1.782894730740056, median 1.3253416246657592, std: 1.5363588317900598

Residuals

Reprojection error (cam0) [px]: mean 0.1700993760450453, median 0.12631178660313505, std: 0.1588506636665267
Reprojection error (cam1) [px]: mean 0.16657500500086836, median 0.12496487519285435, std: 0.15793375682231509
Gyroscope error (imu0) [rad/s]: mean 0.005267476816980646, median 0.004248795664264001, std: 0.003903541885363387
Accelerometer error (imu0) [m/s^2]: mean 0.02275587528247855, median 0.016915922290629458, std: 0.019609228387161764

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99976766 0.01935367 0.00948985 0.02309192]
[-0.01931065 0.99980293 -0.00460423 -0.00172745]
[-0.00957709 0.00441991 0.99994437 -0.01373189]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99976766 -0.01931065 -0.00957709 -0.02325142]
[0.01935367 0.99980293 0.00441991 0.00134089]
[0.00948985 -0.00460423 0.99994437 0.01350403]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.009571161293213361

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.99975549 0.01948706 0.01045099 -0.0719989]
[-0.01944045 0.9998007 -0.00454259 -0.00171177]
[-0.01053743 0.00433831 0.99993507 -0.013929]
[0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.99975549 -0.01944045 -0.01053743 0.07180124]
[0.01948706 0.9998007   0.00433831 0.0031749]
[0.01045099 -0.00454259 0.99993507 0.01467278]
[0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.009604980913365741

Baselines:

Baseline (cam0 to cam1):
[[0.99999953 0.00012917 0.00096179 -0.09507737]
[-0.00012923 0.99999999 0.00006287 0.00001952]
[-0.00096178 -0.00006299 0.99999954 -0.00017502]
[0.   0.   0.   1.   ]]
baseline norm: 0.09507753639648628 [m]

Gravity vector in target coords: [m/s^2]
[-0.39261651 -9.22580991 -3.30131892]

Calibration configuration
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cam0

Camera model: pinhole
Focal length: [430.72820455489654, 431.7138091373145]
Principal point: [431.2946883153815, 246.80296682738822]
Distortion model: radtan
Distortion coefficients: [0.0018285002925504943, -0.0005471378306267974, 5.4402114498175235e-05, 0.0010243936658000717]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.022 [m]
Spacing 0.006599999999999999 [m]

cam1

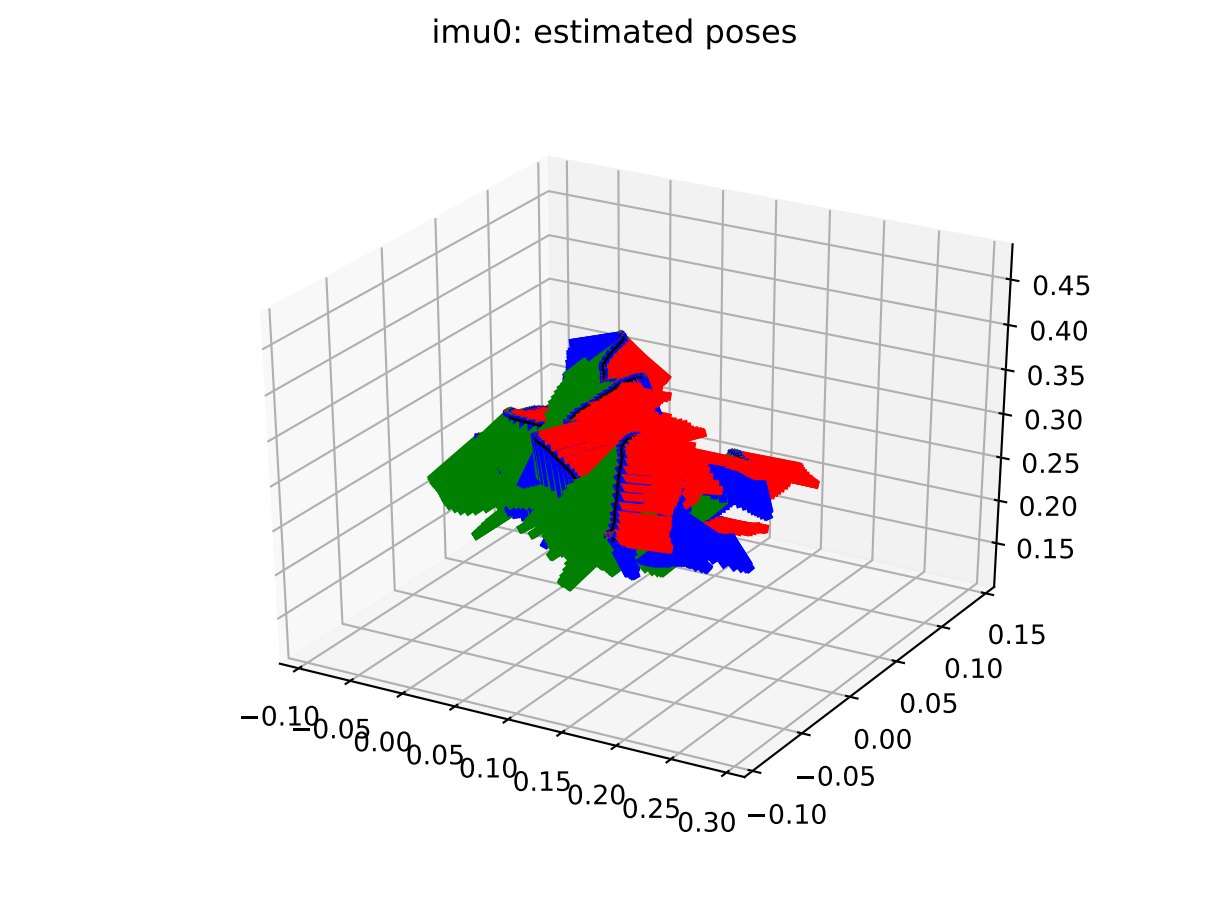
Camera model: pinhole
Focal length: [430.6645601012484, 431.576638780907]
Principal point: [430.8367088763259, 246.88206317251186]
Distortion model: radtan
Distortion coefficients: [0.0016976580217912742, -0.0008304597825719565, 4.6874720208806566e-05, 0.0008959416739602189]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.022 [m]
Spacing 0.006599999999999999 [m]

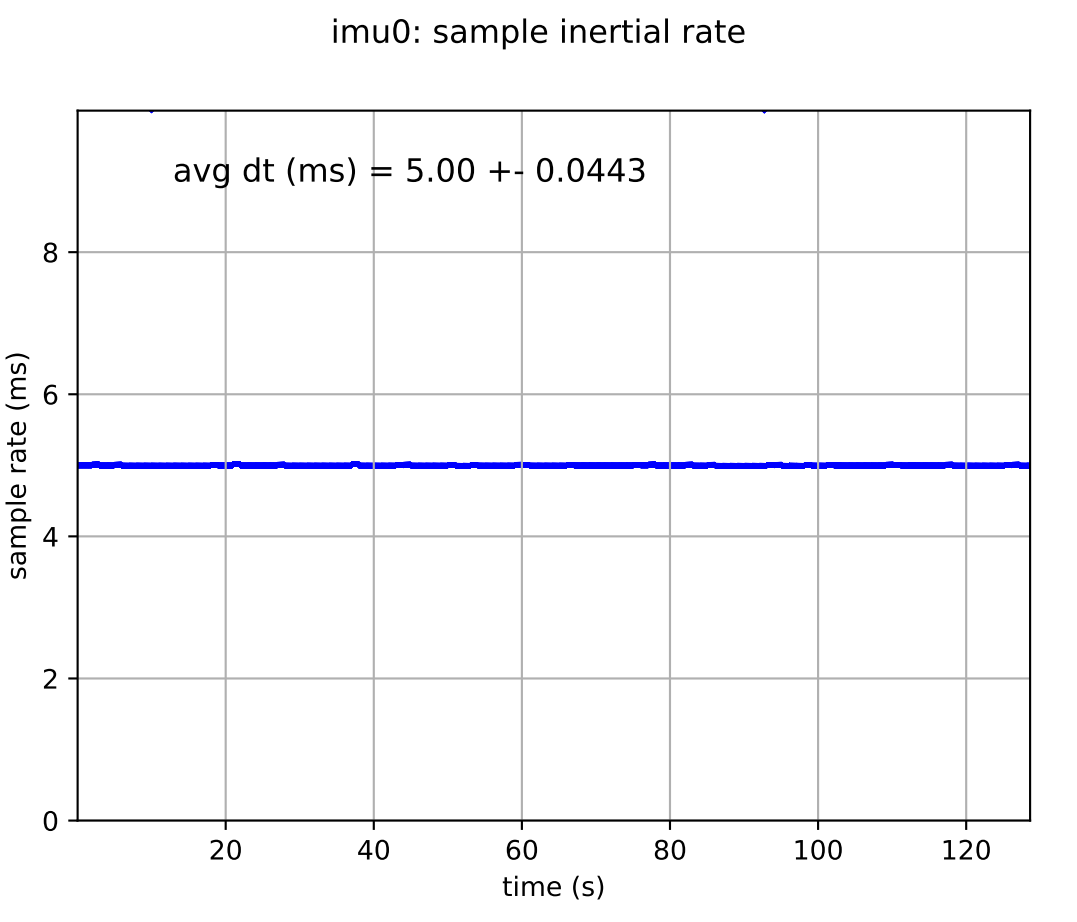
IMU configuration
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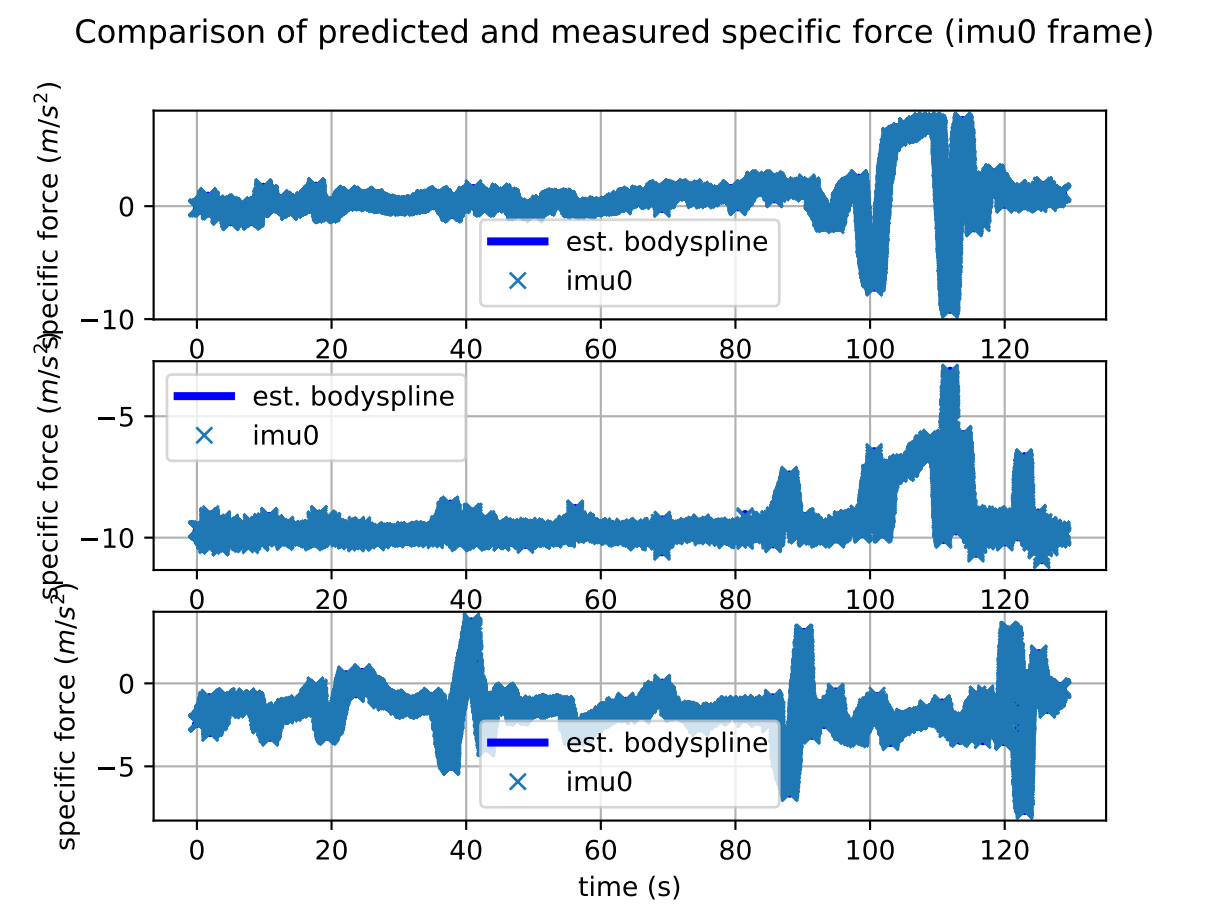
IMU0:

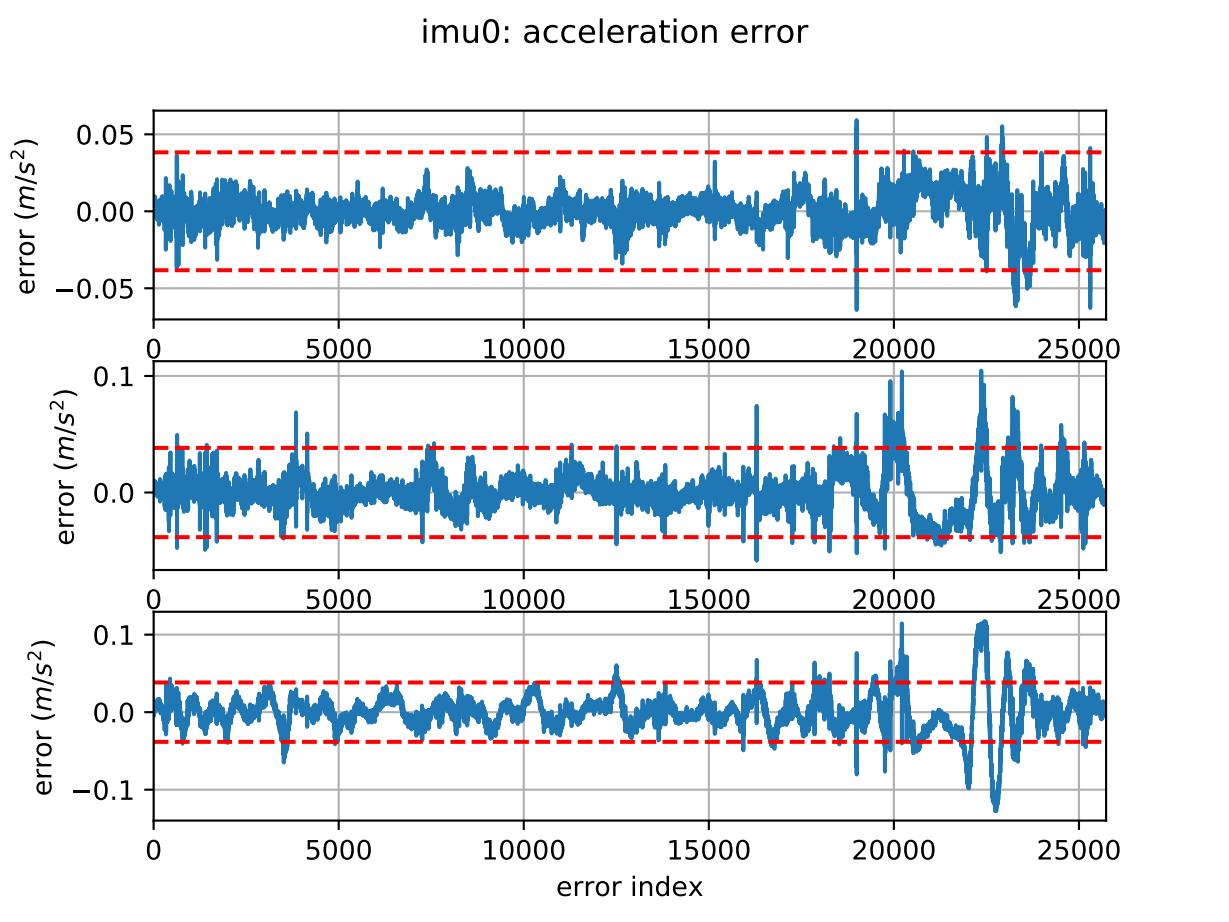
Model: calibrated
Update rate: 200.0

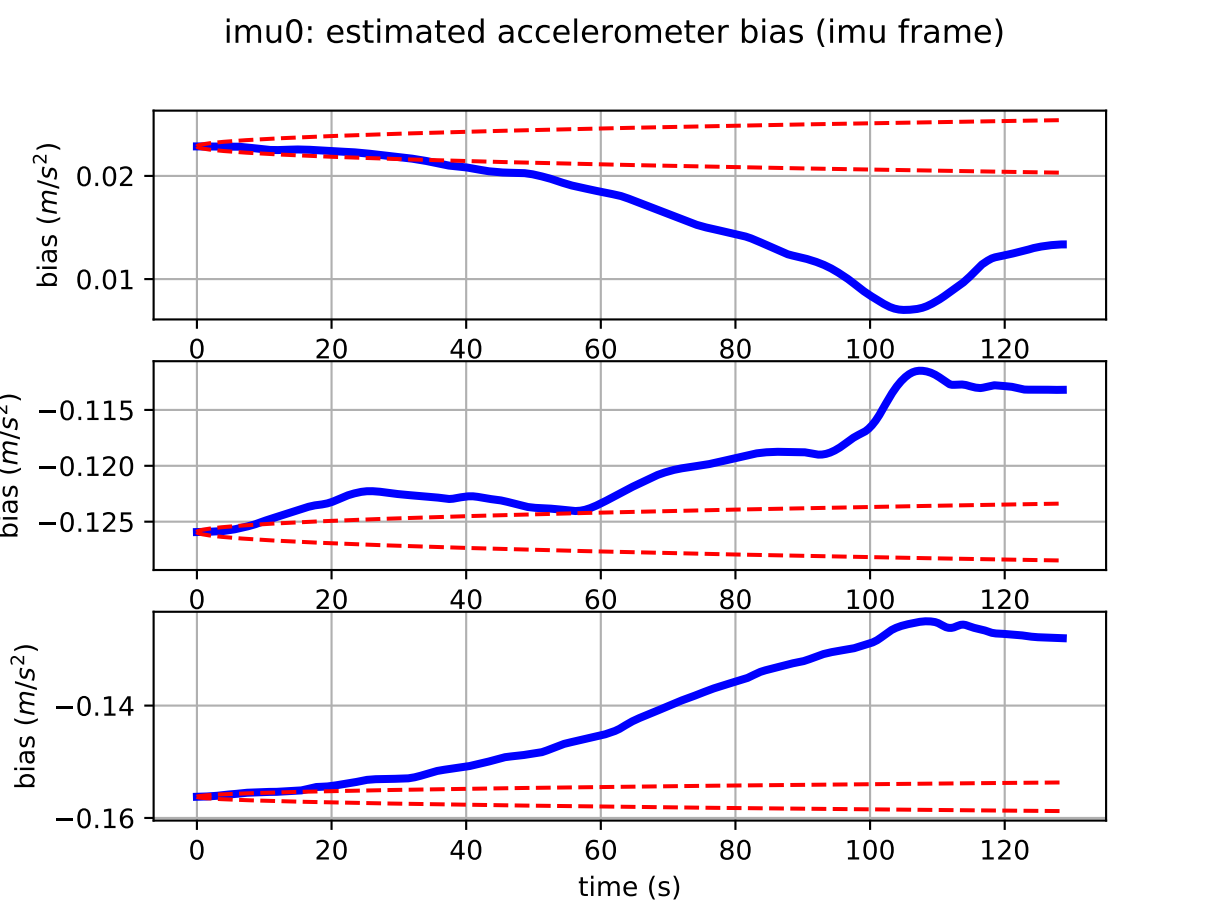
Accelerometer:
Noise density: 0.0009025117101219337
Noise density (discrete): 0.012763443006549741
Random walk: 7.488299992323122e-05
Gyroscope:
Noise density: 0.00014873590113597437
Noise density (discrete): 0.002103443285982787 8
Random walk: 5.161589260762349e-06
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

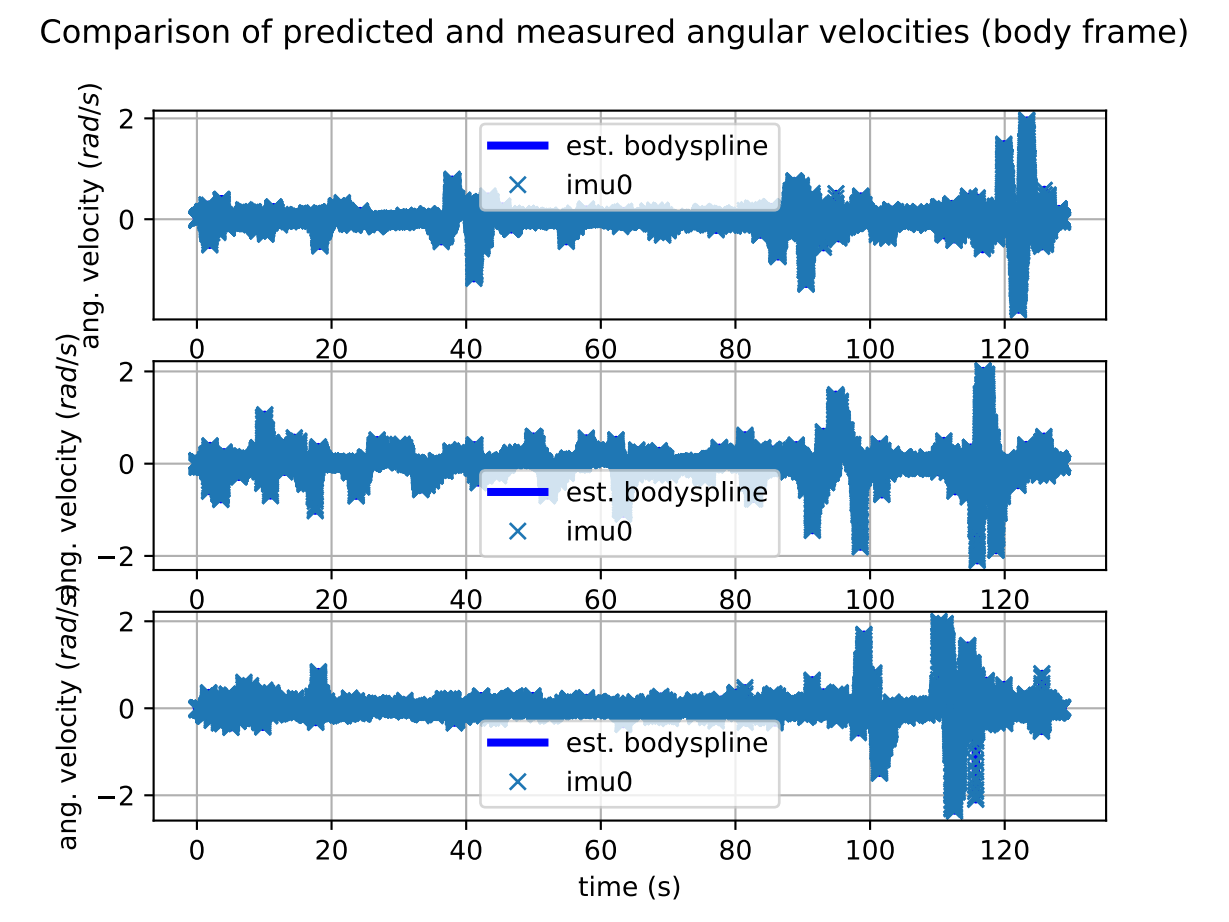

imu0: estimated poses
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30

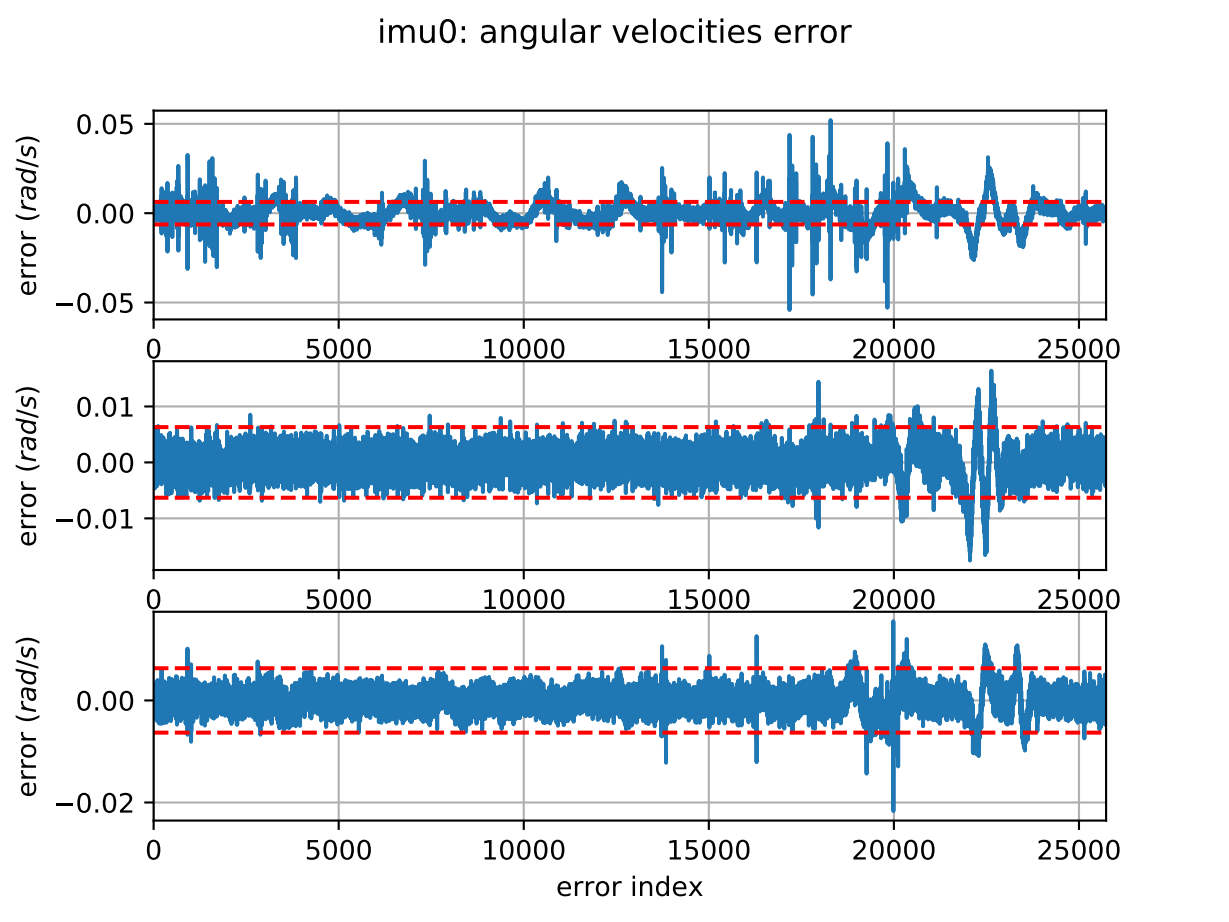

imu0: sample inertial rate
avg dt (ms) = 5.00 +- 0.0443
sample rate (ms)
time (s)
0
2
4
6
8
20
40
60
80
100
120

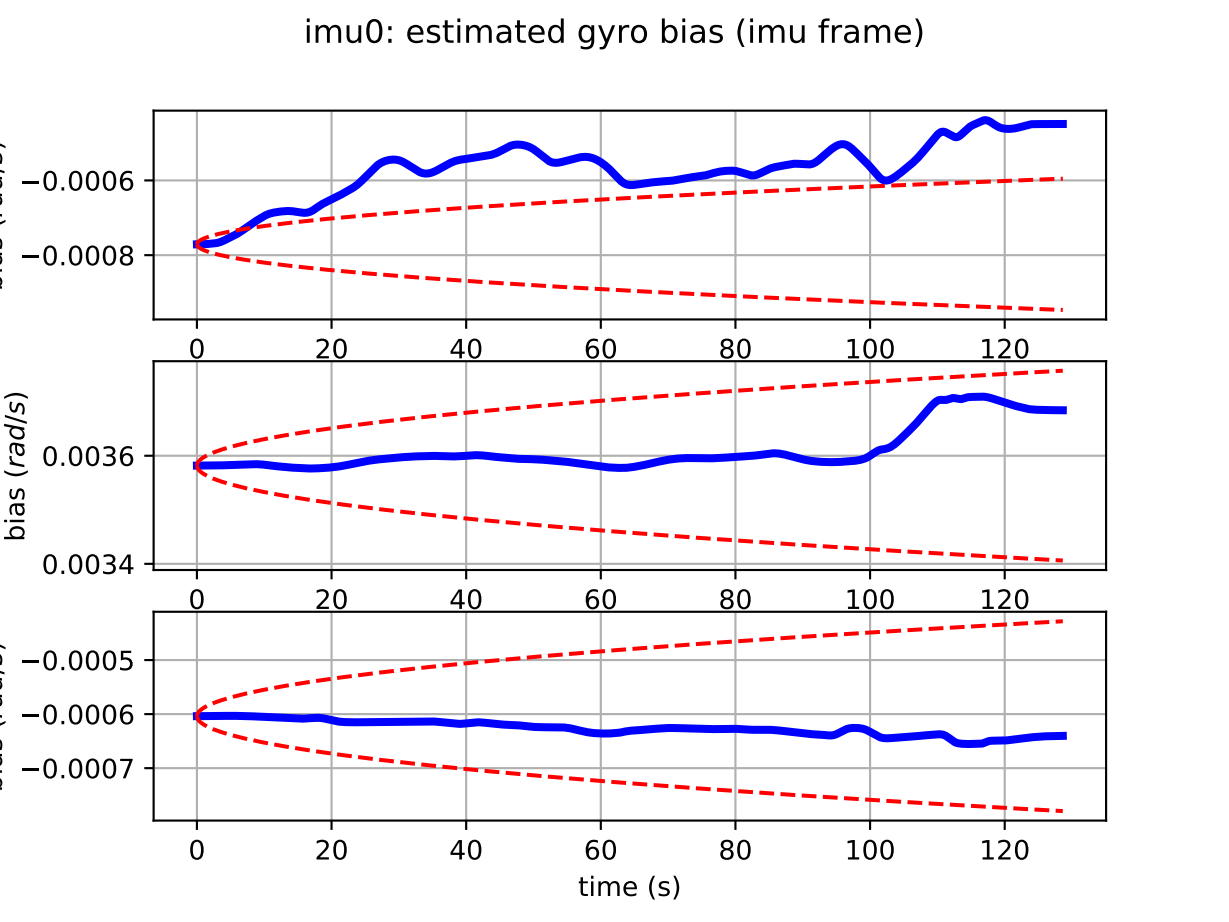

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−10
−5
0
20
40
60
80
100
120
time (s)

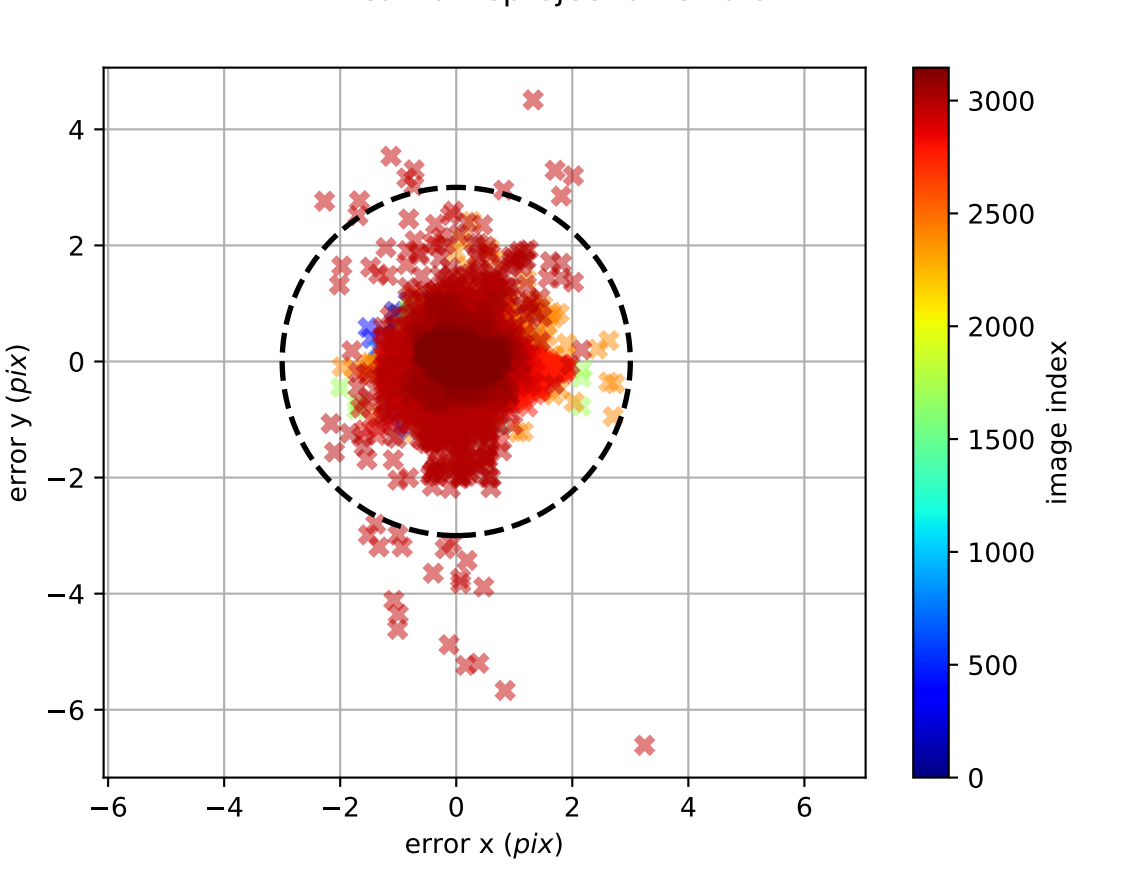

imu0: acceleration error
error (m/s^2)
0.05
0.00
−0.05
0
5000
10000
15000
20000
25000
error (m/s^2)
0.1
0.0
0
5000
10000
15000
20000
25000
error (m/s^2)
0.1
0.0
−0.1
0
5000
10000
15000
20000
25000
error index

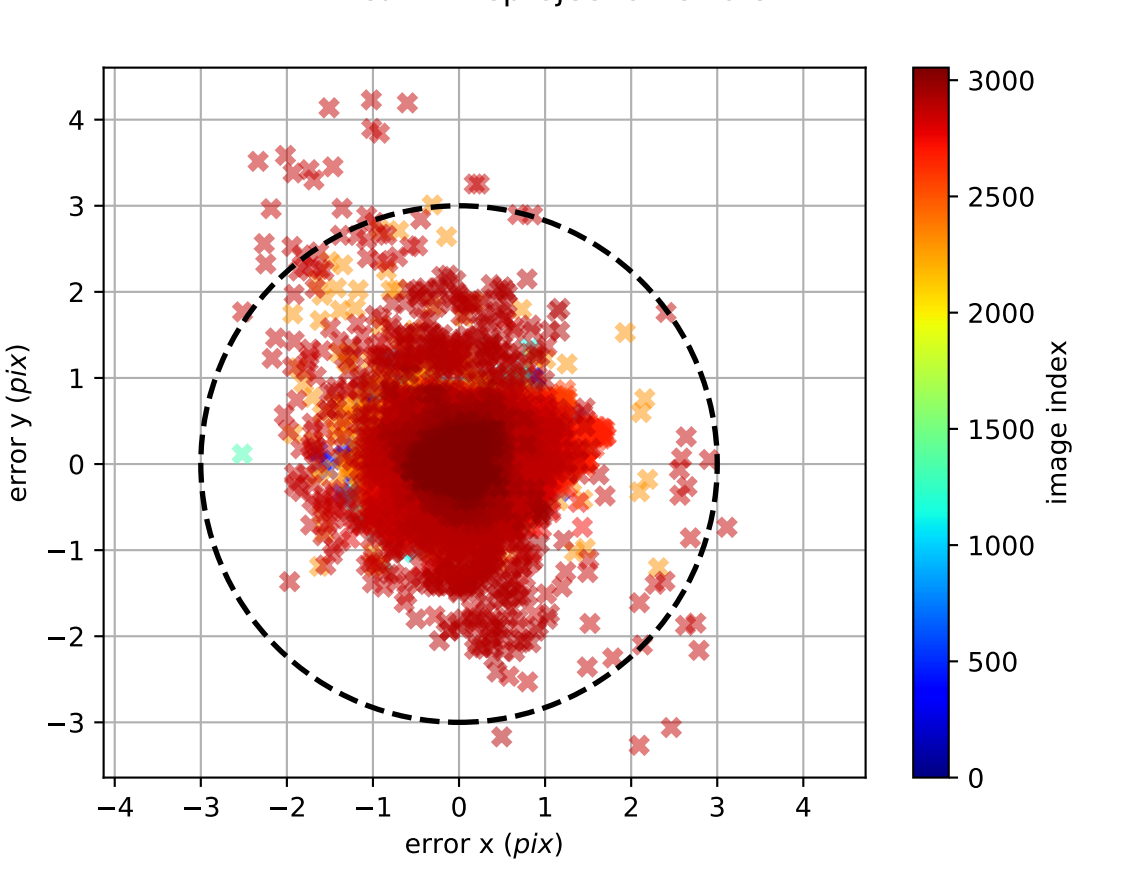

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.02
0.01
0
20
40
60
80
100
120
bias (m/s^2)
−0.115
−0.120
−0.125
0
20
40
60
80
100
120
bias (m/s^2)
−0.14
−0.16
0
20
40
60
80
100
120
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
2
0
−2
0
20
40
60
80
100
120
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
5000
10000
15000
20000
25000
error (rad/s)
0.01
0.00
−0.01
0
5000
10000
15000
20000
25000
error (rad/s)
0.00
−0.02
0
5000
10000
15000
20000
25000
error index


imu0: estimated gyro bias (imu frame)
−0.0006
−0.0008
0
20
40
60
80
100
120
bias (rad/s)
0.0036
0.0034
−0.0005
−0.0006
−0.0007
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−6
−4
−2
0
2
4
6
0
500
1000
1500
2000
2500
3000


cam1: reprojection errors
error x (pix)
error y (pix)
image index
4
3
2
1
0
−1
−2
−3
−4
3000
2500
2000
1500
1000
500
0